§6.2

Comparing Fractions

Five strategies for comparing fractions, and the denseness of rational numbers

Today's Goal

After completing this section, you should be able to:

- Compare two fractions using common denominators, common numerators, missing parts, a benchmark number, or more larger parts versus fewer smaller parts, and choose an efficient strategy for a given pair
- Test a conjecture about comparing fractions against the strategies above
- Find a rational number between any two given rational numbers

Learning Target 14

LT.14 Comparing Fractions

I can apply multiple strategies for comparing fractions.

Which Is Larger?

Which is larger, $\frac{3}{8}$ or $\frac{5}{8}$? Why?

Both fractions have the same denominator, so the whole is divided into the same number of equal parts. $\frac{3}{8}$ is 3 one-eighths and $\frac{5}{8}$ is 5 one-eighths.

The units are the same and 5 is more than 3, so $\frac{5}{8} > \frac{3}{8}$.

Common Denominators

Procedure 6.2.2 Comparing Fractions Using Common Denominators

To compare two fractions using common denominators, do the following.

1. If the two fractions do not have common denominators, then rewrite them as equivalent fractions with the same denominator.
2. Whichever fraction has the larger numerator is the larger fraction.

For example, 4 one-fifths is more than 2 one-fifths because the units (i.e., one-fifth) are the same and 4 is more than 2.



Compare Using Common Denominators

Use common denominators to compare each pair of fractions to determine which is larger.

(a) $\frac{3}{8}$ and $\frac{5}{8}$

Same denominator and $3 < 5$.

So $\frac{3}{8} < \frac{5}{8}$.

(b) $\frac{3}{4}$ and $\frac{7}{12}$

$\frac{3}{4} = \frac{3 \cdot 3}{4 \cdot 3} = \frac{9}{12}$

$9 > 7$, so $\frac{9}{12} > \frac{7}{12}$.

So $\frac{3}{4} > \frac{7}{12}$.

(c) $\frac{2}{3}$ and $\frac{3}{5}$

The denominators 3 and 5 have a least common multiple of 15.

$\frac{2}{3} = \frac{2 \cdot 5}{3 \cdot 5} = \frac{10}{15}$

$\frac{3}{5} = \frac{3 \cdot 3}{5 \cdot 3} = \frac{9}{15}$

$10 > 9$, so $\frac{10}{15} > \frac{9}{15}$.

So $\frac{2}{3} > \frac{3}{5}$.

Common Numerators

Procedure 6.2.5 Comparing Fractions Using Common Numerators

To compare two fractions using common numerators, do the following.

1. If the two fractions do not have the same numerator, then rewrite them as equivalent fractions with the same numerator.
2. Whichever fraction has the larger denominator is the smaller fraction.

$\frac{3}{8}$ is 3 one-eighths and $\frac{3}{4}$ is 3 one-fourths. The unit fraction one-eighth is smaller than the unit fraction one-fourth.



Compare Using Common Numerators

Use common numerators to compare each pair of fractions to determine which is larger.

(a) $\frac{3}{5}$ and $\frac{3}{8}$

Same numerator and $8 > 5$.

So $\frac{3}{8} < \frac{3}{5}$.

(b) $\frac{3}{4}$ and $\frac{6}{11}$

$$\frac{3}{4} = \frac{3 \cdot 2}{4 \cdot 2} = \frac{6}{8}$$

$8 < 11$, so $\frac{6}{8} > \frac{6}{11}$.

So $\frac{3}{4} > \frac{6}{11}$.

(c) $\frac{3}{7}$ and $\frac{2}{5}$

The numerators 3 and 2 have a least common multiple of 6.

$$\frac{3}{7} = \frac{3 \cdot 2}{7 \cdot 2} = \frac{6}{14}$$

$$\frac{2}{5} = \frac{2 \cdot 3}{5 \cdot 3} = \frac{6}{15}$$

$14 < 15$, so $\frac{6}{14} > \frac{6}{15}$.

So $\frac{3}{7} > \frac{2}{5}$.

Missing Parts

Procedure 6.2.8 Comparing Fractions Using Missing Parts

To compare two fractions using missing parts, do the following.

1. Determine what each fraction is missing to make one whole.
2. Whichever fraction is missing more is the smaller fraction.

This strategy is useful when fractions are close to one whole and the missing parts are easier to compare than the original fractions.



Compare Using Missing Parts

Use missing parts to compare each pair of fractions to determine which is larger.

(a) $\frac{5}{6}$ and $\frac{6}{7}$

$\frac{5}{6}$ is missing $\frac{1}{6}$ to make a whole.

$\frac{6}{7}$ is missing $\frac{1}{7}$ to make a whole.

Since $\frac{1}{6} > \frac{1}{7}$, we know that $\frac{5}{6}$ is missing more than $\frac{6}{7}$ is missing.

So $\frac{5}{6} < \frac{6}{7}$.

(b) $\frac{5}{8}$ and $\frac{7}{10}$

$\frac{5}{8}$ is missing $\frac{3}{8}$ to make a whole.

$\frac{7}{10}$ is missing $\frac{3}{10}$ to make a whole.

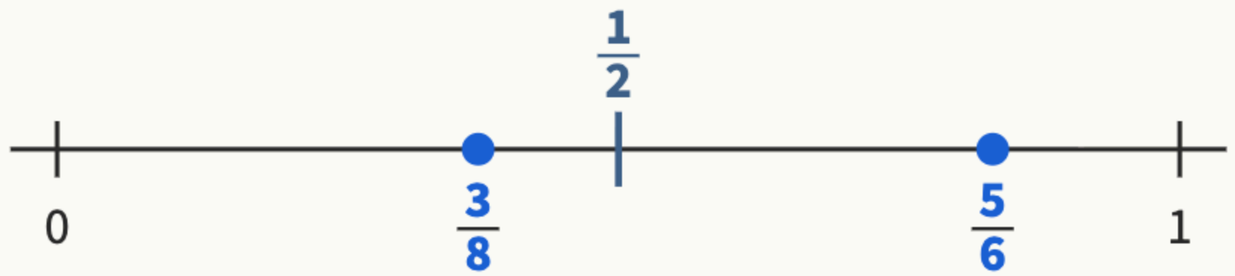
Since $\frac{3}{8} > \frac{3}{10}$ (we get this by using the common numerator strategy), we know that $\frac{5}{8}$ is missing more than $\frac{7}{10}$ is missing.

So $\frac{5}{8} < \frac{7}{10}$.

Comparing to Benchmark Numbers

In some cases, we can compare the two fractions to another fraction, which we call a benchmark number. If one of the fractions is smaller than the benchmark number and the other fraction is larger than the benchmark number, then we know which of the two fractions is larger.

For example, $\frac{3}{8}$ is smaller than $\frac{1}{2}$ and $\frac{5}{6}$ is larger than $\frac{1}{2}$. So $\frac{3}{8} < \frac{5}{6}$.



This strategy is useful when an appropriate benchmark number is easily identified.

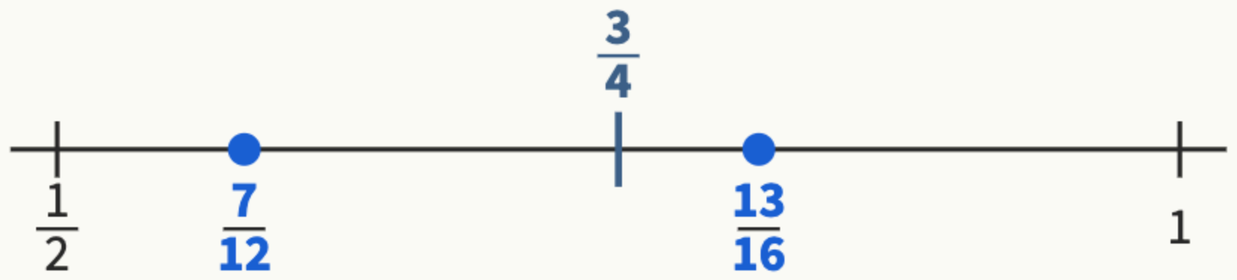
Compare Using a Benchmark Number

Use a benchmark number to compare $\frac{13}{16}$ and $\frac{7}{12}$ to determine which is larger.

Note that half of 16 is 8 and half of 12 is 6. This tells us that $\frac{13}{16}$ and $\frac{7}{12}$ are both larger than $\frac{1}{2}$. So $\frac{1}{2}$ cannot be used as a benchmark number to help us.

Instead, we will use $\frac{3}{4}$ as our benchmark number. We can write $\frac{3}{4}$ as equivalent fractions $\frac{12}{16} = \frac{9}{12}$.

Since $\frac{13}{16} > \frac{12}{16}$ and $\frac{7}{12} < \frac{9}{12}$, we now know that $\frac{13}{16} > \frac{7}{12}$.



Our Last Strategy

Our last strategy is the most straightforward one. If one fraction has a larger unit fraction than the other (i.e., its equal parts are larger) and that same fraction has more parts than the other fraction, then that fraction must be larger.

For example, suppose we are comparing $\frac{3}{7}$ and $\frac{4}{5}$. Note that the unit fraction for $\frac{3}{7}$ is one-seventh and the unit fraction for $\frac{4}{5}$ is one-fifth. So $\frac{4}{5}$ has the larger unit fraction. $\frac{4}{5}$ also has more parts than $\frac{3}{7}$ ($4 > 3$). Therefore, $\frac{4}{5} > \frac{3}{7}$.

This makes sense if you think about comparing distances. 4 miles is larger than 3 feet because miles are larger units than feet and we have more miles than we have feet.

Who Wins?



Compare Using More Larger Parts

Use the "more larger parts vs fewer smaller parts" strategy to compare $\frac{4}{7}$ and $\frac{2}{9}$.

The unit fraction for $\frac{4}{7}$ is one-seventh and the unit fraction for $\frac{2}{9}$ is one-ninth. So $\frac{4}{7}$ has the larger unit fraction (i.e., larger parts).

$\frac{4}{7}$ also has more parts than $\frac{2}{9}$ ($4 > 2$). So $\frac{4}{7} > \frac{2}{9}$.

Choose an Efficient Strategy

Work with your group. For each pair, choose an efficient strategy, then decide which fraction is larger.

(a) $\frac{9}{10}$ and $\frac{11}{12}$ (b) $\frac{3}{8}$ and $\frac{4}{7}$ (c) $\frac{5}{9}$ and $\frac{5}{12}$

The five strategies:

- common denominators
- common numerators
- missing parts
- comparing to benchmark numbers
- more larger parts vs fewer smaller parts

Which Strategy Did You Pick?

Which strategy did your group pick? Did another group pick a different one?

(a) Missing parts. $\frac{9}{10}$ is missing $\frac{1}{10}$ and $\frac{11}{12}$ is missing $\frac{1}{12}$. Since $\frac{1}{10} > \frac{1}{12}$, $\frac{9}{10} < \frac{11}{12}$.

(b) Benchmark number $\frac{1}{2}$. $\frac{3}{8} < \frac{1}{2}$ and $\frac{4}{7} > \frac{1}{2}$, so $\frac{3}{8} < \frac{4}{7}$. More larger parts vs fewer smaller parts also works: one-seventh is larger than one-eighth, and $4 > 3$.

(c) Common numerators. Same numerator and $12 > 9$, so $\frac{5}{12} < \frac{5}{9}$.

The Denseness of Rational Numbers

The rational numbers have a special property that is not true of the integers.

Theorem 6.2.14 Denseness Property of Rational Numbers

Given any two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$, there is another rational number between these two numbers.

Notice that for integers, this doesn't work. For example, there is no integer between 4 and 5.

Find a Rational Number

Find a rational number between $\frac{2}{3}$ and $\frac{3}{4}$.

Rewrite each fraction with the same denominator of 12: $\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}$ and $\frac{3}{4} = \frac{3 \cdot 3}{4 \cdot 3} = \frac{9}{12}$.

There isn't enough space between $\frac{8}{12}$ and $\frac{9}{12}$ to fit in another one-twelfth. So we will rewrite each fraction again with a larger common denominator of 24.

$\frac{2}{3} = \frac{2 \cdot 8}{3 \cdot 8} = \frac{16}{24}$ and $\frac{3}{4} = \frac{3 \cdot 6}{4 \cdot 6} = \frac{18}{24}$.

Now, we can easily choose $\frac{17}{24}$ and be confident that it is between $\frac{2}{3}$ and $\frac{3}{4}$.